Name- Ayush Singh

Course Name - Data Analytics With GenAI

Subject - Tableau

Topic - Tableau Basics

Contact Number - 7054891639

Email ID - as764994@gmail.com

Tableau Workbook - Link

Q1. Explain what Tableau is and why it is preferred over traditional spreadsheet tools for data analysis. Also mention any two industries where Tableau is commonly used and explain one use case.

Tableau is a visual analytics platform that allows users to connect to various data sources and create interactive, shareable dashboards and reports without writing any code. It was developed to make data analysis accessible to everyone, not just technical professionals. You simply drag and drop fields to build charts, maps, and trend lines in seconds.

Traditional tools like Excel are great for storing and calculating data, but they struggle when the dataset gets large, the visualizations need to be interactive, or multiple people need to collaborate on the same view in real time. Tableau handles datasets with millions of rows smoothly, offers built-in chart recommendations, and allows anyone to filter and explore data on their own without touching the underlying numbers. This makes it far more powerful for actual analysis and storytelling with data.

Two industries where Tableau is widely used are **Retail** and **Healthcare**.

In retail, for example, a chain store uses Tableau to track daily sales performance across all store locations on a single dashboard. Store managers can filter by region, product category, or time period and instantly see which stores are underperforming, allowing quick corrective action without waiting for a weekly Excel report.

Q2. After connecting to the Sample – Superstore dataset:

a) **List the main types of fields available (examples: time-based, geographical, numerical).**
b) **Identify any four Dimensions and any four Measures from the dataset.**

a) Main types of fields available:

The dataset contains three broad types of fields — time-based fields like Order Date and Ship Date, geographical fields like State, City, Region, and Country, and numerical fields like Sales, Profit, Quantity, and Discount. There are also categorical text fields like Category, Sub-Category, Segment, and Ship Mode.

b) Four Dimensions and four Measures:

Dimensions (blue fields — qualitative/categorical):

- Category

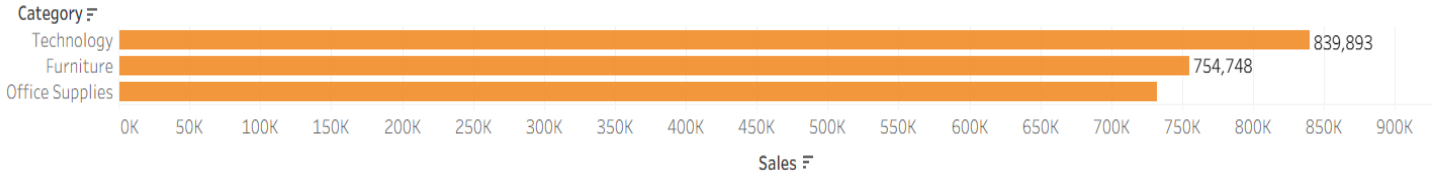
- Sub-Category
- Region
- Customer Name

Measures (green fields — quantitative/numerical):

- Sales
- Profit
- Quantity
- Discount

Q3. Create a Bar Chart showing Total Sales by Category.
Explain:

- **Which field goes to Rows**
- **Which field goes to Columns**
- **Which aggregation is applied to Sales**

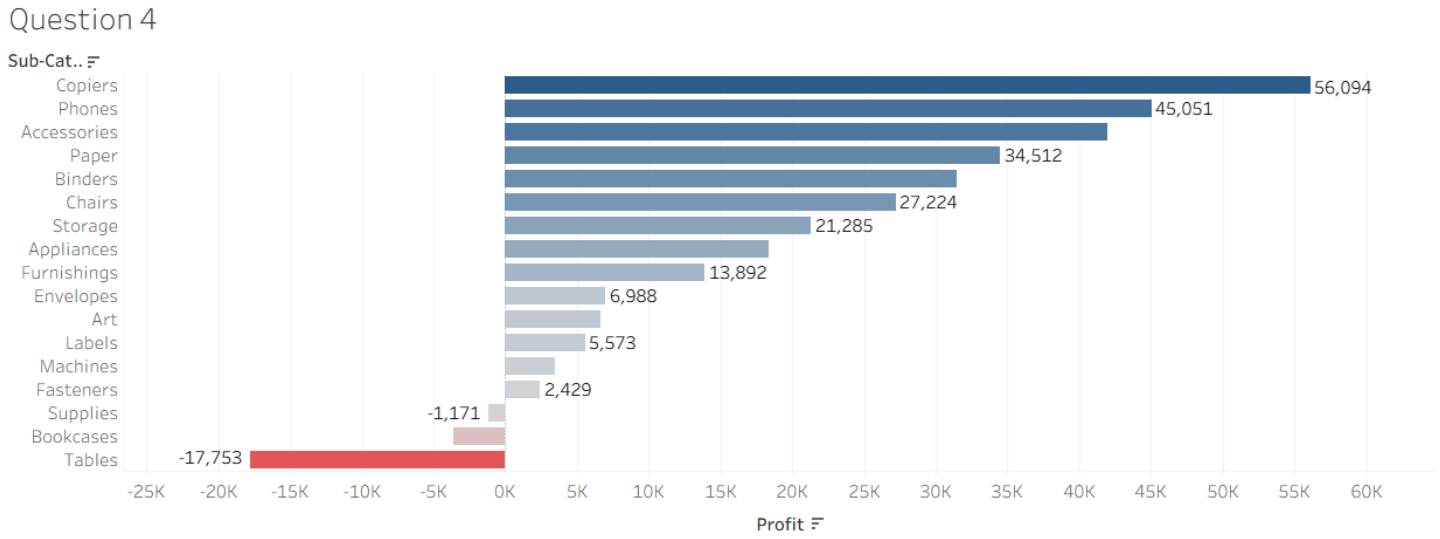


To create this chart, you drag **Category** to the **Rows** shelf and **Sales** to the **Columns** shelf. Tableau automatically applies a **SUM aggregation** to Sales since it is a measure, displaying the total sales for each category.
So to answer specifically:

- **Rows shelf** gets: Category
- **Columns shelf** gets: Sales (SUM)
- **Aggregation applied**: SUM(Sales)

Once these are placed, Tableau defaults to a horizontal bar chart. You can sort it in descending order to immediately see which category leads in sales.
Based on the actual dataset, **Technology** has the highest total sales at approximately **$839,893**, followed by **Furniture** at around **$754,748**, and **Office Supplies** at around **$731,893**.

Q4. What insight can be gained by comparing Sales and Profit together? Then explain how you would create a chart to analyze Profit by Sub-Category.



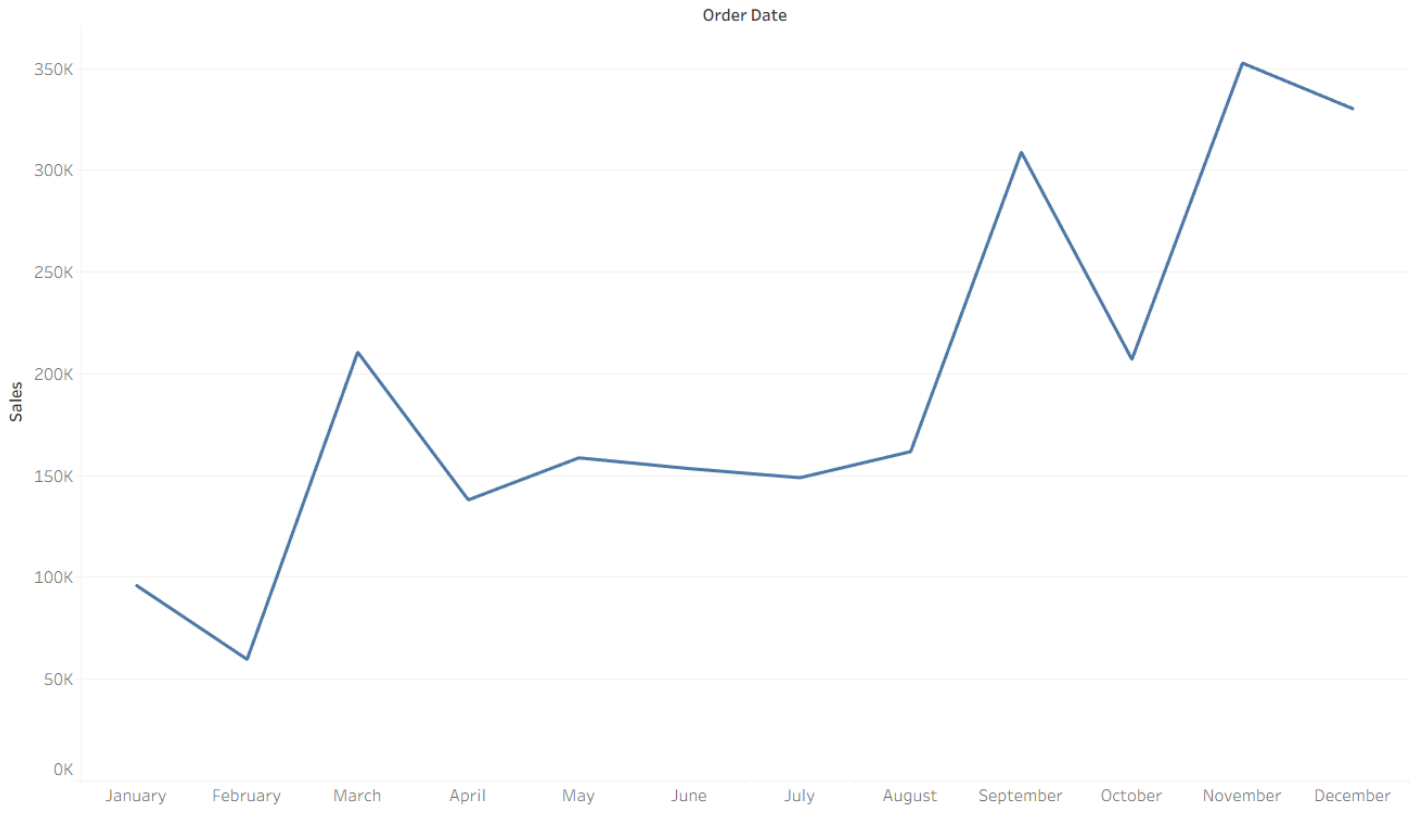
Comparing Sales and Profit together reveals something very important — a category with high sales does not always mean high profit. For example, Furniture has the second highest sales in the Superstore dataset, but its profit is extremely low compared to Technology. This tells a manager that Furniture might have high discounts or high costs eating into margins, and the business should investigate pricing or supplier costs for that category.

To create a **Profit by Sub-Category** chart, you drag **Sub-Category** to the Rows shelf and **Profit** (SUM) to the Columns shelf. You can then sort it to see which sub-categories are profitable and which ones are actually losing money.

From the actual data, **Tables** is the worst performing sub-category with a loss of around **-$17,753**, followed by **Bookcases** at **-$3,632**. On the other end, **Copiers** generate the highest profit at approximately **$56,094**. This kind of insight helps the business decide which products to promote and which to reconsider.

Q5. Create a Line Chart to show Sales Trend over Order Date.

- **Which date level would you choose (Year / Quarter / Month)?**
- **Why are line charts suitable for time-based analysis?**



To create this, you drag **Order Date** to the Columns shelf and **Sales** to the Rows shelf. By default, Tableau shows the year level. You can click the small plus icon on the Order Date pill to drill down to **Quarter** or **Month**.

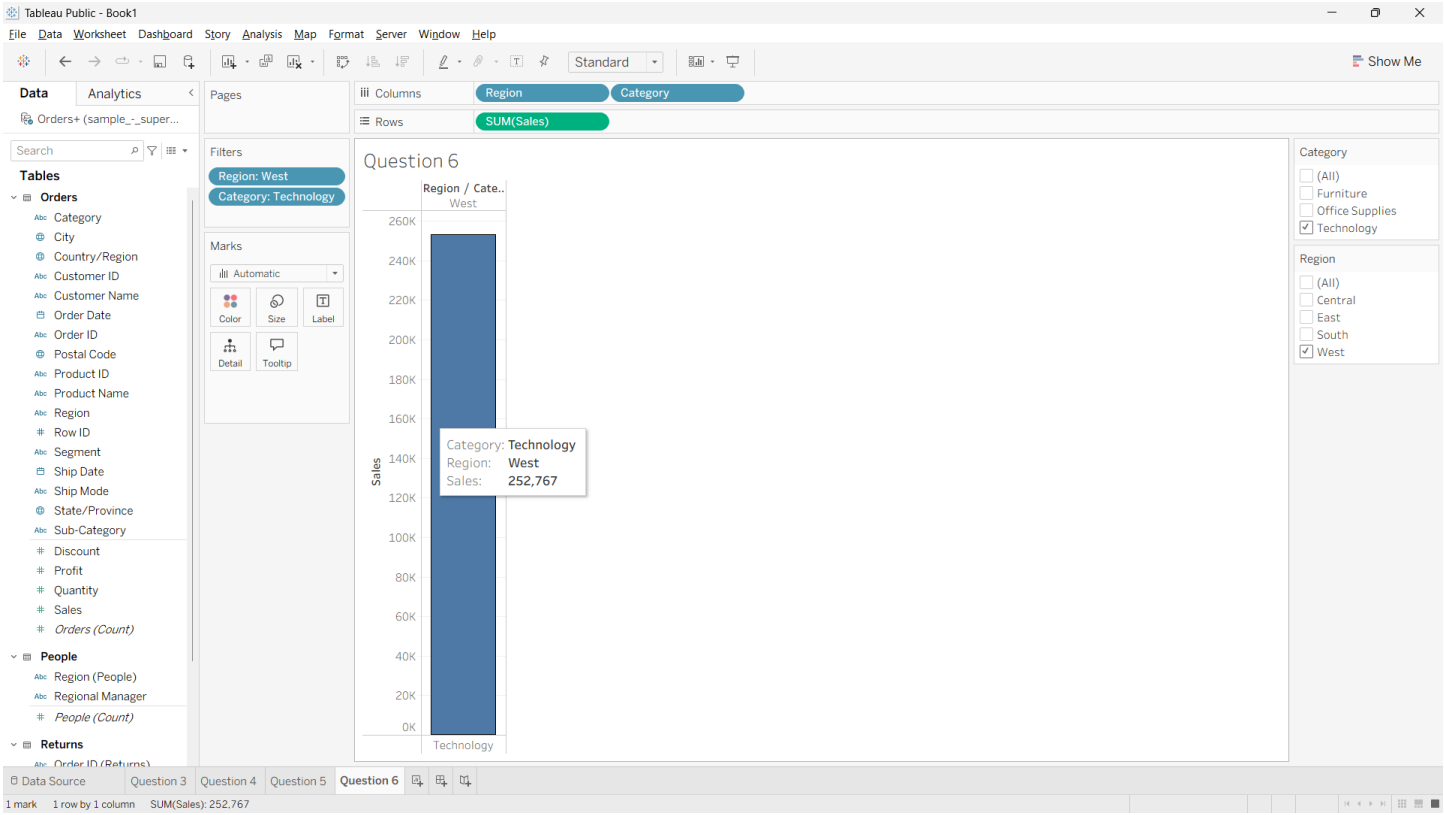
For this analysis, **Month** is the best date level to choose because it shows enough granularity to spot seasonal patterns — like whether sales spike during the holiday months — without being so detailed that the chart becomes noisy.

Line charts are ideal for time-based analysis because they naturally show **continuity and direction**. When you connect data points across time, trends, peaks, dips, and seasonal patterns become immediately visible. A bar chart at the monthly level would technically show the same numbers, but a line makes it much easier for the eye to follow the movement over time and draw conclusions about growth or decline.

Q6. Apply a Basic Filter to display data only for:

- **Region = "West"**
- **Category = "Technology"**

Explain the steps and mention which shelf is used.



The shelf used for this is called the **Filters shelf**, located in the top-left area of the worksheet view.

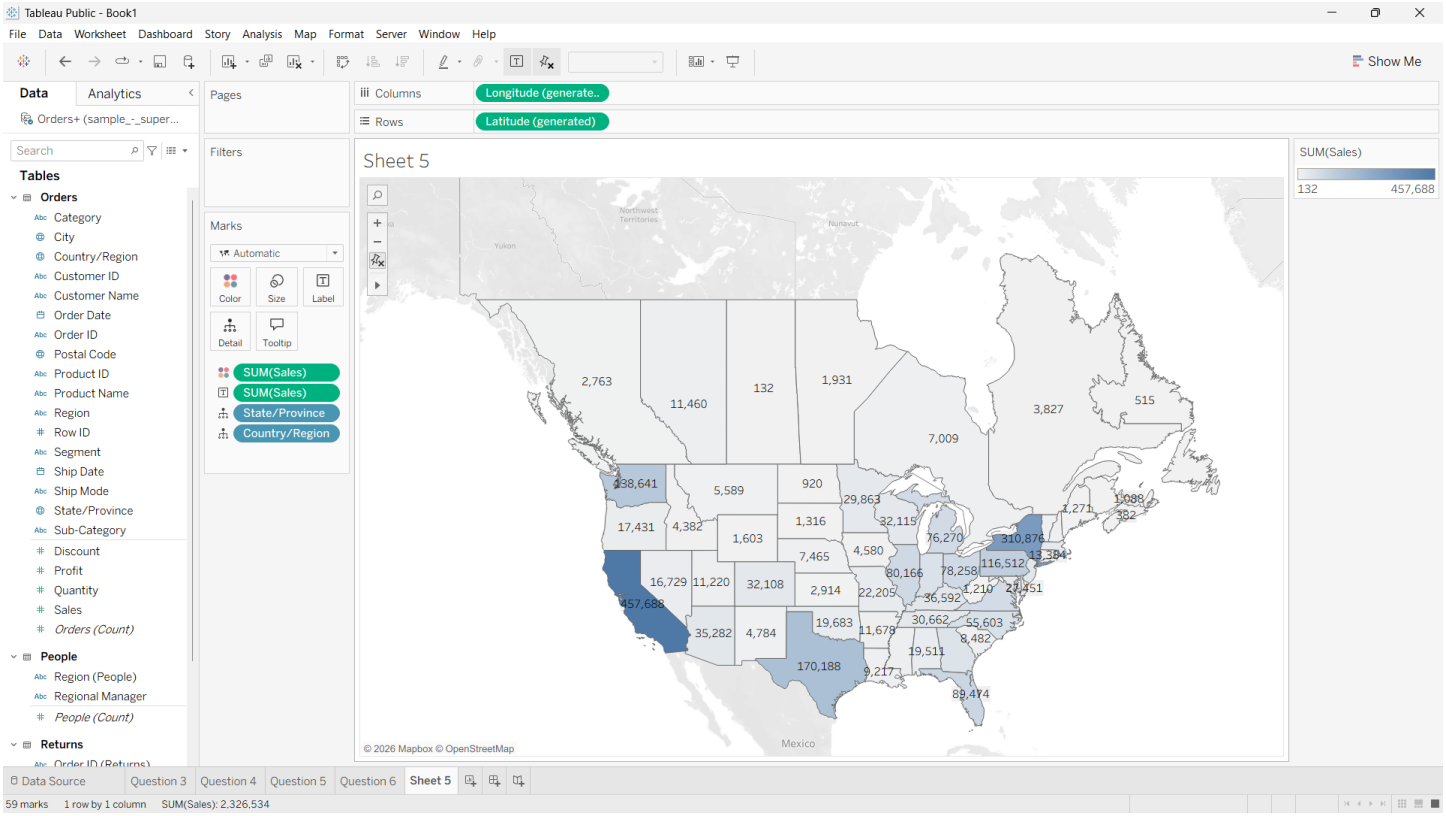
Steps: First, locate the **Region** field in the Data pane on the left side. Drag it directly onto the **Filters shelf**. A dialog box will appear showing all available region values — Central, East, South, and West. Select only **West** and click OK.

Next, find the **Category** field and drag it to the **Filters shelf** as well. From the dialog, select only **Technology** and click OK.

After both filters are applied, the entire view automatically updates to show only data where the region is West and the category is Technology. You will notice the number of records drops significantly, and all charts on that worksheet now reflect only this filtered subset.

You can also right-click either filter pill and select **Show Filter** to make it appear as an interactive dropdown on the view, which is useful for dashboards.

Q7. Explain the business importance of geographical analysis. Then describe how you would create a Map showing Sales by State using Sample – Superstore.



Geographical analysis is important because performance almost always varies by location. A business cannot treat every region the same — some states might be highly profitable while others barely break even. Without a map view, these differences are buried in rows and columns. A geographic visualization makes regional patterns immediately obvious, helping management decide where to invest in marketing, where to open new stores, or where to investigate operational problems.

In the Superstore dataset, we can already see that the **East** and **West** regions have far higher profits than the **Central** region, which signals that strategies working in the coasts may not be translating to the middle of the country.

To create a **Sales by State map** in Tableau, drag the **State/Province** field onto the canvas or double-click it. Tableau automatically recognizes it as a geographic field and switches to a map view. Then drag **Sales** to the **Color** card in the Marks pane. Tableau fills each state with a color shade — darker meaning higher sales. You can also drag Sales to the **Label** card to show exact numbers on the map. This gives a clear visual of which states are driving the most revenue at a single glance.

Q8. What is the difference between a Worksheet and a Dashboard in Tableau? Why is a dashboard more useful for decision-makers?

A **Worksheet** is a single canvas where you build one chart or visualization at a time. Every chart — whether a bar chart, line graph, or map — lives in its own worksheet. It is the basic building block of Tableau.

A **Dashboard** is a combined view where you bring multiple worksheets together on one screen. You can arrange charts side by side, add filters that work across all charts at once, and create a complete story from multiple angles on a single page.

For decision-makers, a dashboard is far more useful because they rarely need to look at just one chart. A manager wants to see sales trends, regional performance, and category-wise profit all at the same time without switching between tabs. A dashboard gives them that complete picture in one view. They can also interact with it — clicking on one chart can automatically filter all other charts on the same dashboard, which makes exploration fast and intuitive. It saves time and supports faster, better-informed decisions.

Q9. What is Tableau Prep, and why is it important in real-world data projects? What is Tableau Prep, and why is it important in real-world data projects? Then explain how you would:

- **Remove null values**
- **Remove duplicate records**
- **Change incorrect data types**

Tableau Prep is a data preparation tool by Tableau that allows analysts to clean, reshape, and combine data before analyzing it in Tableau Desktop. In real-world projects, raw data is almost never clean — it comes with missing values, duplicates, wrong data types, inconsistent formatting, and data spread across multiple files. Tableau Prep provides a visual, flow-based interface to handle all of this without writing SQL or Python.

It is important because dirty data leads to wrong insights. If you analyze data with null values or duplicate rows, your totals will be incorrect, and any decision made on that analysis could be harmful to the business.

How to remove null values in Tableau Prep: Open the dataset in Tableau Prep and click on a field in the profile view. Null values appear clearly as a separate bar labeled "Null." You can right-click on the null bar and select **Exclude** to remove those rows, or you can use a **Clean Step** and filter out rows where that field is null.

How to remove duplicate records: Add a **Clean Step** to your flow. In the toolbar, look for the option to remove duplicates. Tableau Prep can identify rows that are identical across

all fields and keep only one copy. Alternatively, you can group rows by a unique identifier like Order ID and remove extras.

How to change incorrect data types: In the profile view at the top of each field, you'll see an icon indicating the current data type — Abc for text, # for number, calendar for date. If a date column is being read as text, simply click that icon and select the correct data type from the dropdown. Tableau Prep will attempt to convert the values automatically and flag any that it cannot convert.

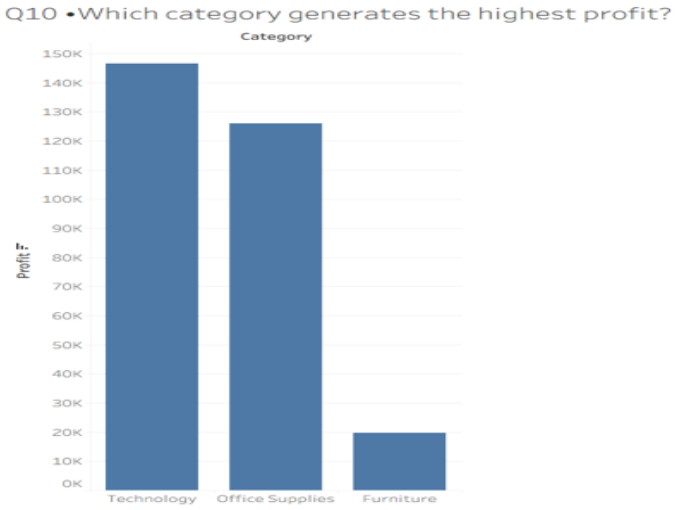
Q10. A retail manager wants answers to the following:

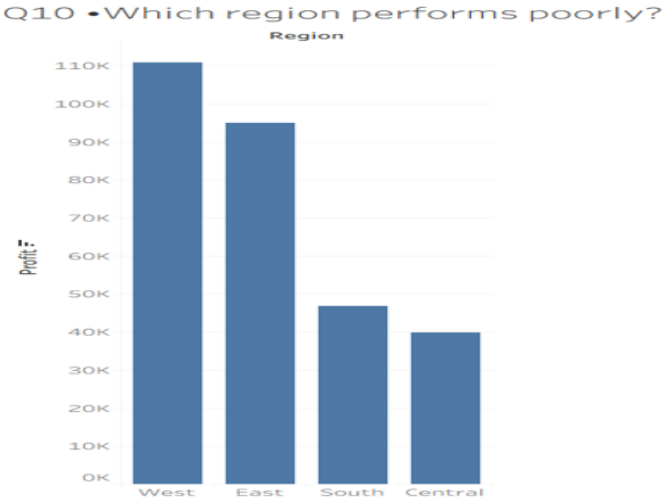
- **Which category generates the highest profit?**
- **Which region performs poorly?**
- **How does sales change over time?**

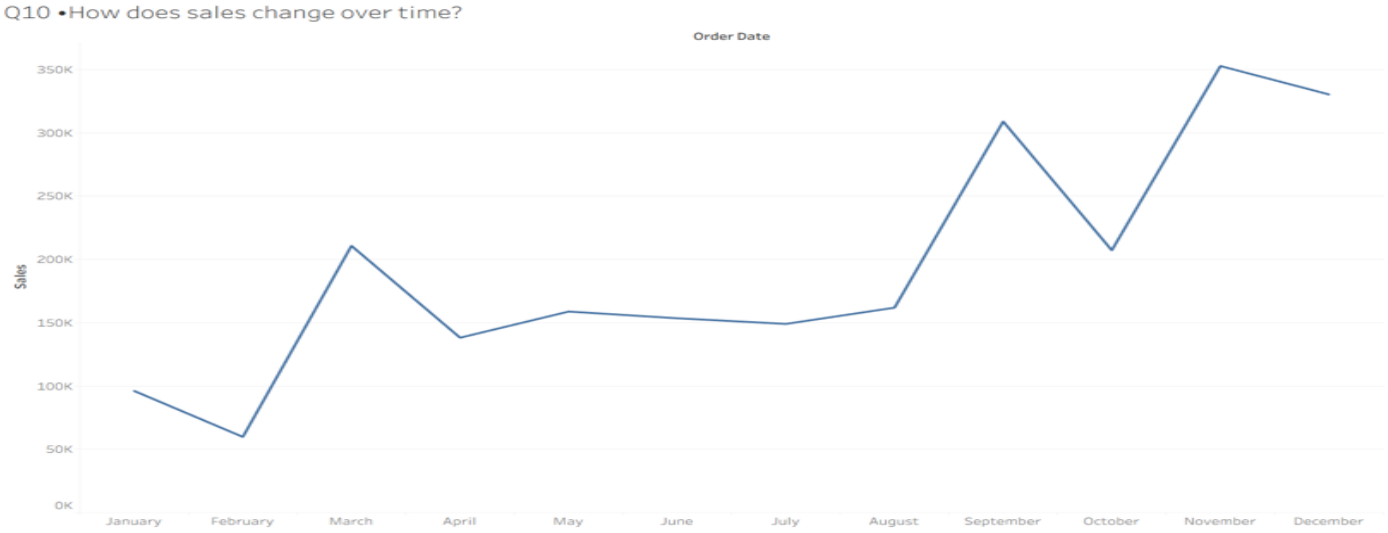
a) Mention which charts you would use

b) Identify one key insight that management could derive

c) Explain how this helps in business decision-making







a) Charts to use:

To find which category generates the highest profit — a **Bar Chart** with Category on one axis and SUM(Profit) on the other, sorted in descending order, is the best choice.

To find which region performs poorly — again a **Bar Chart** with Region and Profit.

To see how sales change over time — a **Line Chart** with Order Date on the x-axis and SUM(Sales) on the y-axis, broken down by month or quarter.

b) One key insight management could derive:

Looking at the actual Superstore data, the most striking insight is that **Furniture has the second highest sales but by far the lowest profit** among all three categories — only about $19,730 in profit compared to Technology's $146,543. Meanwhile, the **Central region has the lowest profit** at around $39,865, nearly three times less than the West region's $110,799. This combination tells management that Furniture sold in the Central region is likely a significant drag on overall business profitability.

c) How this helps in business decision-making:

This kind of analysis directly informs action. If management sees that Furniture has high sales but poor profit, they know to investigate — are discounts too aggressive? Are shipping costs too high for bulky items? Is the supplier pricing unfavorable? They can then adjust discount policies or renegotiate with suppliers.

For the underperforming Central region, management can dig deeper to see which states or cities are pulling the numbers down, then decide whether to run targeted promotions, change the product mix, or reallocate sales team resources. The line chart showing how sales change over time helps with planning — if there's a consistent sales dip in Q1 every year, inventory and staffing can be adjusted accordingly. Data-driven decisions like these reduce guesswork and help the business allocate resources where they will have the most impact.